

Software Requirements Specification (SRS)

Project: Online Bookstore System (Buying & Selling Platform) **Version:** 1.0 (Approved Final)

Authors:

Team 4

(PRUTHVIRAJ H PES2UG24CS381, PUNEET LINGASHETTAR PES2UG24CS382, PUNITH K PES2UG24CS383, PURVITHA MACHIREDDY PES2UG24CS384)

Date: September 4, 2026

Revision history

Version	Date	Author	Change summary	Approval
0.1	18-08-2025	Team 4	Initial draft specification for Online Bookstore.	Draft
0.5	01-09-2026	Team 4	Updated architectural stack to React, Django, and PostgreSQL. Added seller feature requirements.	Reviewed
1.0	04-09-2026	Team 4	Final baseline release. Fully expanded Functional Requirements (20 FRs), NFRs (6 NFRs), Security Objectives (2), and Security Requirements (5 SRs) with RTM.	Approved

Approvals

Role	Name	Signature / Status	Date
Course Evaluator / Coordinator	Department Faculty Coordinator	SUBMITTED	04-09-2026

1. Introduction

1.1 Purpose

This document serves as the formal Software Requirements Specification (SRS) for the Online Bookstore system—an integrated, full-stack e-commerce solution enabling users to buy new and pre-owned books online as well as list their own books for sale. It outlines the complete functional capabilities, non-functional performance benchmarks, software/hardware interface definitions, and security specifications. This document guides the development, verification, and evaluation teams throughout the project lifecycle.

1.2 Scope

The Online Bookstore system encompasses dual operational workflows: buyer operations (user account creation, product discovery, advanced filtering, cart management, checkout, online payment, order tracking, and peer reviews) and seller/admin operations (user book listing submissions, catalog management, inventory adjustments, order status updates, review moderation, and sales analytics reporting). The system integrates external third-party services including payment gateways (e.g., Razorpay/Stripe APIs) and transactional email dispatchers (e.g., SMTP/SendGrid). Physical warehousing, courier logistics handling, and raw card processing are explicitly excluded from the software boundary.

1.3 Audience

This specification is structured for Project Evaluators, Software Engineers, Front-End (React) Developers, Back-End (Django) Developers, QA/Test Automation Engineers, and System Administrators.

1.4 Definitions & Acronyms

- **FR:** Functional Requirement
- **NFR:** Non-Functional Requirement
- **SR:** Security Requirement
- **React:** Open-source front-end JavaScript library for building component-based user interfaces.
- **Django / DRF:** Python-based high-level web framework and Django REST Framework for API construction.
- **PostgreSQL:** Enterprise-grade open-source relational database management system.
- **JWT:** JSON Web Token used for stateless, secure HTTP session authentication.
- **RBAC:** Role-Based Access Control enforcing permission levels (Guest, Customer/Seller, Admin).
- **PCI-DSS:** Payment Card Industry Data Security Standard.
- **RTM:** Requirements Traceability Matrix linking requirements to design modules and test cases.

2. Overall Description

2.1 Product Perspective

The Online Bookstore operates as an independent web application based on a decoupled client-server architecture. The user interface is driven by a React SPA (Single Page Application) communicating over secure HTTPS JSON REST APIs with a Django backend server. System data—comprising user profiles, book listings, inventories, order records, payment audit trails, and product reviews—persists inside a relational PostgreSQL database. Integration points include external payment gateway processors and transactional email notification engines.

2.2 Major Product Functions

- **Identity & Account Governance:** Secure user registration, email verification, login with JWT tokens, password reset, and profile management.
- **Catalog Discovery & Search:** Real-time keyword search (title, author, ISBN, genre), dynamic multi-attribute filtering, and detailed product display.
- **Peer-to-Peer Selling / Book Listing:** Capability for registered users/sellers to upload secondhand/new books with custom descriptions, condition ratings, pricing, and uploaded cover photos.
- **Cart & Order Processing:** Real-time cart calculation, shipping address validation, wishlist preservation, and secure checkout.
- **Payment Gateway Integration:** Automated tokenized payment processing through PCI-DSS compliant third-party gateways.
- **Order Fulfillment & Tracking:** Lifecycle tracking (Pending, Processing, Shipped, Delivered, Cancelled) with automated email notifications.
- **Reviews & Moderation:** Verified buyer rating system with admin moderation to remove inappropriate content.
- **Admin Control & Analytics:** Centralized dashboard for catalog curation, inventory thresholds, user management, and aggregate sales reporting.

2.3 User Roles & Characteristics

- **Guest (Unregistered Visitor):** Can search, filter, and view book details and reviews. Must register to buy or sell books.
- **Customer / Buyer:** Registered user who browses books, manages cart/wishlist, places orders via payment gateway, tracks orders, and writes reviews.
- **Seller / Book Provider:** Registered user who lists books for sale, inputs book details and physical condition, sets pricing, and tracks submitted items.
- **Administrator:** Internal power user with full rights to oversee the platform, curate catalog entries, approve/reject seller submissions, update order statuses, moderate reviews, and inspect sales metrics.

2.4 Operating Environment

- **Client Environment:** Cross-platform modern web browsers (Google Chrome v100+, Mozilla Firefox v100+, Apple Safari v15+, Microsoft Edge) on desktop, tablet, and mobile viewports.
- **Application Server:** Django REST Framework hosted on Linux containerized

infrastructure (Docker / AWS EC2 / Render) running Python 3.11+.

- **Database Tier:** PostgreSQL 15+ relational database server with connection pooling and automated daily backup routines.

2.5 Constraints

- **Technology Stack Enforcement:** Mandatory use of React.js for frontend, Django REST framework for backend services, and PostgreSQL for database persistence.
- **Data Protection & Card Security:** Strictly zero storage of plain-text credit/debit card numbers (PAN) or card verification values (CVV) on local databases. All financial transactions must use tokenization via PCI-DSS compliant gateways.
- **Team & Schedule Constraints:** Project development, testing, and documentation restricted to a 4-person engineering team within the designated academic term.

3. External Interface Requirements

3.1 User Interfaces

The front-end user interface built using React components provides intuitive, clean layout navigation. Key UI screens include: Homepage with featured and top-selling books, Catalog & Advanced Filter View, Detailed Product Screen with condition notes, Cart & Multi-Step Checkout Flow, Customer Order Dashboard, Seller Book Submission Form, and Admin Management Console. Interfaces adhere to basic accessibility standards including WCAG 2.1 AA readable contrast, focus states for keyboard navigation, and explicit alt attributes for cover art images.

3.2 Hardware Interfaces

The application is entirely web-based and requires no specialized physical hardware. Standard client computing devices (desktops, laptops, tablets, smartphones) equipped with standard network connectivity and web browsers serve as client nodes.

3.3 Software Interfaces

- **Payment Gateway API:** RESTful API integration with Stripe / Razorpay for tokenized checkout, Webhook event handling for payment status callbacks.
- **Transactional Email API:** SMTP / SendGrid API integration for automatic dispatch of registration tokens, password resets, and order confirmation receipts.
- **Database Interface:** Django Object-Relational Mapper (ORM) interfacing directly with PostgreSQL via psycopg2 database drivers over secure local/VPC sockets.

3.4 Communications Interfaces

- **HTTP/HTTPS:** Mandatory TLS 1.2/1.3 encryption across all REST API communications between React frontend and Django backend.
- **JSON Data Transfer:** Standardized JSON payload structures for requests and responses across all HTTP endpoints.
- **Webhooks:** Secure cryptographic signature verification for incoming payment gateway asynchronous event notifications.

4. System Features (Detailed Functional Requirements)

4.1 User Accounts & Authentication Module

Req ID	Requirement Description	Priority	Acceptance Criteria / Test Ref	Dependencies
BOOK-F-001	The system shall allow new users to register using full name, email address, and a secure password with mandatory email activation.	High	Account is created in 'Inactive' state; activation email sent; account becomes 'Active' upon token verification. (Test: TC-Auth-01)	Requires SendGrid/SMTTP API integration.
BOOK-F-002	The system shall authenticate registered users using email and password, returning a signed JWT access and refresh token pair upon success.	High	Valid credentials yield a 200 OK with JWT token; invalid credentials return a 401 Unauthorized status. (Test: TC-Auth-02)	Core authentication API endpoint.
BOOK-F-003	The system shall allow users to request password recovery via a time-sensitive emailed reset link expiring after 30 minutes.	High	Reset link generated with cryptographically secure token; password updated successfully within 30 min window. (Test: TC-Auth-03)	Requires email dispatcher service.
BOOK-F-004	The system shall allow authenticated users to view and update their profile details including phone number and default shipping addresses.	Medium	Profile edits persist instantly in PostgreSQL and reflect in subsequent session loads. (Test: TC-Auth-04)	User profile Django model.
BOOK-F-005	The system shall enforce server-side Role-Based Access Control (RBAC) categorizing users as Customer/Buyer,	High	Requests to admin/seller API endpoints by customer JWT tokens are blocked with HTTP 403 Forbidden. (Test: TC-Auth-05)	Django DRF permission classes.

Req ID	Requirement Description	Priority	Acceptance Criteria / Test Ref	Dependencies
	Seller, or Admin.			

4.2 Search, Catalog & Book Discovery Module

Req ID	Requirement Description	Priority	Acceptance Criteria / Test Ref	Dependencies
BOOK-F-006	The system shall enable users to execute real-time text searches across book titles, authors, ISBN numbers, and genre tags.	High	Sub-string search returns matching book list within 2 seconds for a database of 10,000 records. (Test: TC-Cat-01)	Indexed PostgreSQL text fields.
BOOK-F-007	The system shall allow users to filter catalog results by price range, book condition (New/Used), rating, and sort by price or popularity.	Medium	Catalog view immediately updates displaying only books matching selected filter criteria. (Test: TC-Cat-02)	Django REST filtering backends.
BOOK-F-008	The system shall render a detailed book view page displaying high-res cover art, full description, publisher, stock level, price, seller info, and reviews.	High	Page displays complete metadata for valid book ID; returns 404 for invalid ID. (Test: TC-Cat-03)	React detail view component.
BOOK-F-009	The system shall allow authenticated customers to add items to a persistent digital shopping cart, update quantities, or remove items.	High	Cart total, item subtotals, and taxes automatically recalculate upon quantity change. (Test: TC-Cat-04)	PostgreSQL Cart model.
BOOK-F-010	The system shall allow logged-in customers to save books to a personal wishlist for deferred purchase.	Low	Wishlist items persist across logout/login sessions and can be directly moved to cart. (Test: TC-Cat-05)	Wishlist relationship schema.

4.3 Seller Book Listing Module

Req ID	Requirement Description	Priority	Acceptance Criteria / Test Ref	Dependencies
BOOK-F-011	The system shall allow authenticated users/sellers to post a book for sale by uploading book details, condition rating (New/Like New/Good/Acceptable), price, stock quantity, and cover photos.	High	Form validates image format and required fields; listing is saved in 'Pending Approval' state. (Test: TC-Sel-01)	Requires image upload/storage handling.

4.4 Checkout, Payment & Order Processing Module

Req ID	Requirement Description	Priority	Acceptance Criteria / Test Ref	Dependencies
BOOK-F-012	The system shall allow customers to initiate checkout, enter or select shipping addresses, and review order totals including taxes and shipping fees.	High	Order summary accurately reflects address, selected items, item quantities, and calculated grand total. (Test: TC-Ord-01)	Cart & Order models.
BOOK-F-013	The system shall integrate with a PCI-DSS compliant third-party payment gateway (Stripe/Razorpay) to process payments securely via tokenization.	High	Successful payment returns gateway transaction ID, converts cart to Order, and clears cart; failed payment alerts user with error message. (Test: TC-Ord-02)	Requires Payment Gateway API key.
BOOK-F-014	The system shall send an automated transactional email receipt with itemized breakdown immediately	Medium	Email dispatched within 60 seconds of successful gateway callback. (Test: TC-Ord-03)	Requires email service.

Req ID	Requirement Description	Priority	Acceptance Criteria / Test Ref	Dependencies
	following payment confirmation.			
BOOK-F-015	The system shall provide real-time order tracking status (Placed, Processing, Shipped, Delivered, Cancelled) on the customer dashboard.	Medium	Dashboard accurately reflects status updates made by administrators or automated webhooks. (Test: TC-Ord-04)	Order status state machine.
BOOK-F-016	The system shall allow customers to view their complete historical order log with itemized invoices and current delivery statuses.	Medium	All past orders associated with the customer account are correctly listed in chronological order. (Test: TC-Ord-05)	PostgreSQL order history queries.

4.5 Administration, Curation & Analytics Module

Req ID	Requirement Description	Priority	Acceptance Criteria / Test Ref	Dependencies
BOOK-F-017	The system shall allow administrators to create, edit, deactivate, or delete master book catalog items and review user-submitted seller listings.	High	Admin approvals publish listings to live catalog; edits reflect immediately across catalog searches. (Test: TC-Adm-01)	Admin RBAC permissions.
BOOK-F-018	The system shall allow administrators to update stock counts, set low-stock thresholds, and transition order fulfillment statuses.	High	Inventory decrements automatically upon order placement; low stock alert flags items with stock < 5 units. (Test: TC-Adm-02)	Inventory management engine.
BOOK-F-019	The system shall permit verified buyers to post a 1-5 star rating and	Medium	Only users with 'Delivered' order status can post reviews; flagged	Review moderation pipeline.

Req ID	Requirement Description	Priority	Acceptance Criteria / Test Ref	Dependencies
	review, while allowing administrators to moderate or delete offensive content.		reviews deleted by admin disappear from product page. (Test: TC-Adm-03)	
BOOK-F-020	The system shall provide administrators with a sales dashboard summarizing total revenue, total orders, top-selling titles, and active seller listings.	Low	Dashboard metrics match exact aggregate calculations executed against the PostgreSQL order database. (Test: TC-Adm-04)	Django aggregate queries & React charts.

5. Non-Functional Requirements (NFRs)

Req ID	Requirement Description	Category	Priority	Acceptance Measurement & Test Ref
BOOK-NF-001	Catalog searches and product detail API endpoints shall respond within \le2.0 seconds for 95% of requests under normal operating conditions.	Performance	High	Load testing via JMeter confirms 95th percentile response time \le2.0s. (Test: TC-Perf-01)
BOOK-NF-002	The web application server and database service shall maintain a operational uptime of \ge99.5\% per calendar month, excluding scheduled maintenance.	Reliability	High	Automated health checks log \ge99.5\% uptime over a 30-day monitoring window. (Test: TC-Rel-01)
BOOK-NF-003	The system shall support at least 500 concurrent active user sessions without exceeding 20% degradation in baseline response	Scalability	Medium	Simulated multi-user load test with 500 concurrent connections passes latency threshold. (Test: TC-Perf-02)

Req ID	Requirement Description	Category	Priority	Acceptance Measurement & Test Ref
	latency.			
BOOK-NF-004	The front-end user interface shall adapt dynamically across desktop (1920x1080), tablet (768x1024), and mobile (375x812) viewports adhering to WCAG 2.1 AA standards.	Usability & Accessibility	High	Automated Lighthouse UX accessibility audit score > 90/100 across 3 device breakpoints. (Test: TC-UX-01)
BOOK-NF-005	All historical order records, financial audit trails, and payment transaction logs shall be retained in PostgreSQL for a minimum of 3 years.	Data Retention	Medium	Archival retrieval test confirms data from 3 years prior can be queried in < 5 seconds. (Test: TC-Dat-01)
BOOK-NF-006	Database transactions for order placement and inventory reduction must adhere to strict ACID compliance to prevent race conditions during concurrent sales.	Data Integrity	High	Simultaneous purchases of the last inventory unit by 2 threads results in exactly 1 successful order and 1 out-of-stock rejection. (Test: TC-Int-01)

5.1 Security Requirements

5.1.1 Security Objectives

- **Confidentiality & Payment Data Isolation:** Ensure all customer credentials, personal profiles, and payment tokens remain strictly protected against unauthorized disclosure both in transit (TLS 1.2/1.3 encryption) and at rest (bcrypt password hashing, no plaintext card data stored).
- **Integrity & Access Authorization:** Prevent unauthorized administrative operations, illegal inventory modifications, or unauthorized seller catalog manipulations by enforcing rigorous server-side Role-Based Access Control (RBAC) and JWT signature verification on every protected endpoint.

5.1.2 Granular Security Requirements

Req ID	Requirement Description	Priority	Acceptance Criteria & Test Case Ref
BOOK-SR-001	User passwords must	High	PostgreSQL

Req ID	Requirement Description	Priority	Acceptance Criteria & Test Case Ref
	be salted and hashed using PBKDF2/bcrypt via Django's default authentication engine. Passwords must never be logged or stored in plain text.		'auth_user' table stores only bcrypt hash strings; application logs contain no plain text passwords. (Test: TC-Sec-01)
BOOK-SR-002	All network transport between the React frontend, Django REST API, and external service providers must be encrypted via HTTPS enforcing TLS 1.2 or higher.	High	SSL Labs vulnerability scan confirms TLS 1.2/1.3 enforcement and rejection of unencrypted HTTP requests. (Test: TC-Sec-02)
BOOK-SR-003	User sessions shall be managed using signed, time-limited JSON Web Tokens (JWT) with 15-minute access expiry and server-side token revocation upon user logout.	High	Expired or blacklisted JWT tokens return HTTP 401 Unauthorized when presented to protected endpoints. (Test: TC-Sec-03)
BOOK-SR-004	The Django API must enforce role-based permission middlewares (IsAdminUser, IsSeller, IsOwner) to prevent vertical/horizontal privilege escalation.	High	Attempts by regular customers to access admin routes (e.g., /api/admin/sales/) are rejected with HTTP 403 Forbidden. (Test: TC-Sec-04)
BOOK-SR-005	The application shall strictly avoid storing Primary Account Numbers (PAN), CVVs, or card expiry dates locally, delegating payment tokenization entirely to PCI-DSS compliant gateways.	High	Database schema review verifies complete absence of PAN/CVV columns; payment processing uses gateway tokens exclusively. (Test: TC-Sec-05)

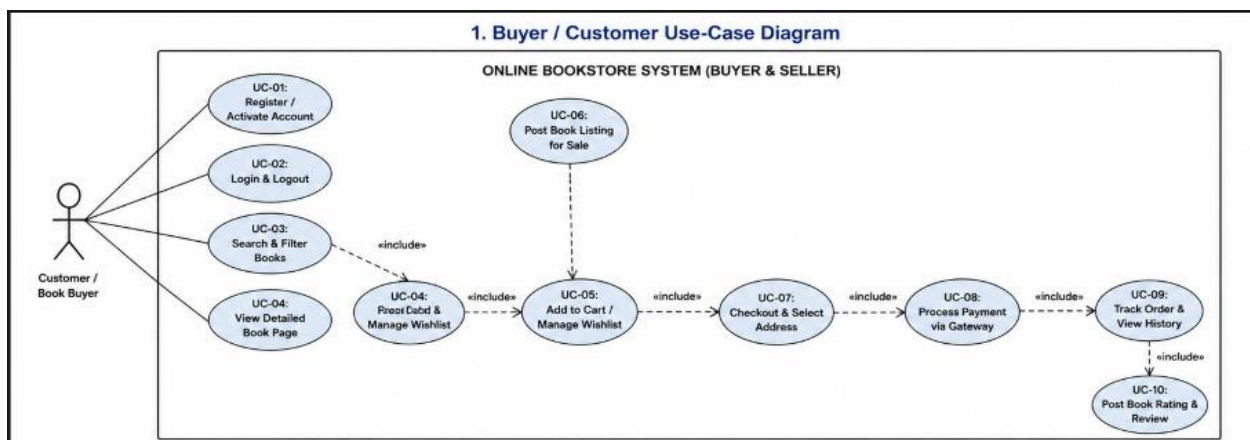
6. Quality Attributes & Acceptance Tests

- **Project Exit Criteria:**
 1. 100% of all 20 Functional Requirements (BOOK-F-001 through BOOK-F-020) implemented, tested, and verified.
 2. All 6 Non-Functional Requirements (BOOK-NF-001 through BOOK-NF-006) validated against performance targets.

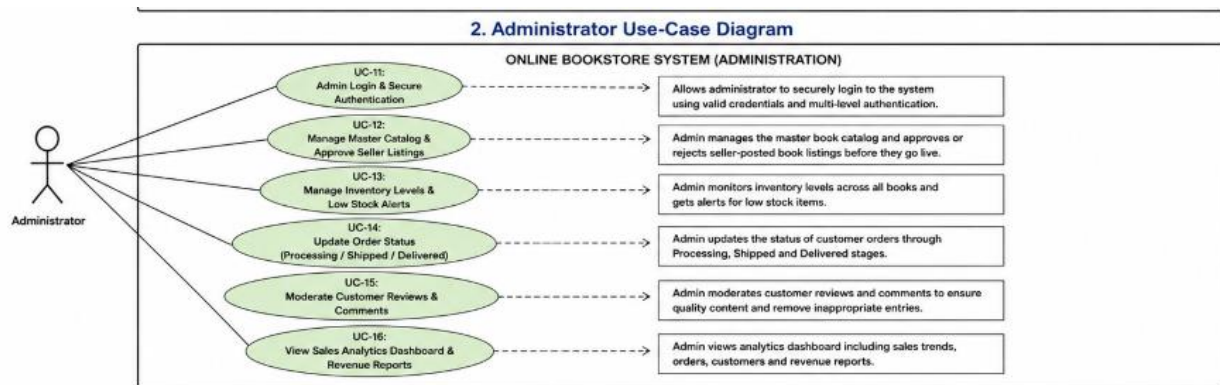
3. All 5 Security Requirements (BOOK-SR-001 through BOOK-SR-005) pass security audit checklists without critical vulnerabilities.
 4. Requirements Traceability Matrix (RTM) updated with all test cases marked as "Approved".
- **Acceptance Test Suites:**
 - *Suite 1 - User Auth & Access Test Suite:* Registration, activation, login, password reset, RBAC verification.
 - *Suite 2 - Catalog & Search Test Suite:* Keyword search, filters, pagination, seller book submission validation.
 - *Suite 3 - Checkout & Payment Test Suite:* Address selection, cart sync, Stripe/Razorpay tokenized checkout, email receipt dispatch.
 - *Suite 4 - Admin & Fulfillment Test Suite:* Catalog approval, inventory thresholds, review moderation, sales dashboard calculations.
 - *Suite 5 - Quality & Performance Test Suite:* JMeter load test (500 users), WCAG 2.1 AA accessibility audit, SSL/TLS security scan.

7. System Models and Diagrams

7.1 Customer & Book Seller Workflows



7.2 Administrator Workflows



8. Requirements Traceability Matrix (RTM)

Req ID	Requirement Summary	Design Module	Test Case Ref	Status	Owner / Assignee
BOOK-F-001	User Registration & Activation	AuthModule	TC-Auth-01	Approved	PRUTHVIRAJ H
BOOK-F-002	JWT Login & Session Auth	AuthModule	TC-Auth-02	Approved	PRUTHVIRAJ H
BOOK-F-003	Emailed Password Reset	AuthModule	TC-Auth-03	Approved	PRUTHVIRAJ H
BOOK-F-004	User Profile & Address Edit	AuthModule	TC-Auth-04	Approved	PRUTHVIRAJ H
BOOK-F-005	RBAC Permission Enforcement	AuthModule	TC-Auth-05	Approved	PRUTHVIRAJ H
BOOK-F-006	Real-time Book Search	CatalogModule	TC-Cat-01	Approved	PUNEET LINGASHETTA R
BOOK-F-007	Catalog Multi-Filter & Sort	CatalogModule	TC-Cat-02	Approved	PUNEET LINGASHETTA R
BOOK-F-008	Detailed Book View Page	CatalogModule	TC-Cat-03	Approved	PUNEET LINGASHETTA R
BOOK-F-009	Cart Management Engine	CatalogModule	TC-Cat-04	Approved	PUNEET LINGASHETTA R
BOOK-F-010	Wishlist Preservation	CatalogModule	TC-Cat-05	Approved	PUNEET LINGASHETTA R
BOOK-F-011	Seller Book Listing Post	CatalogModule	TC-Sel-01	Approved	PUNEET LINGASHETTA R
BOOK-F-012	Checkout & Address Selection	OrderModule	TC-Ord-01	Approved	PUNITH K
BOOK-F-013	Gateway Payment Processing	OrderModule	TC-Ord-02	Approved	PUNITH K
BOOK-F-014	Emailed Order Confirmation	OrderModule	TC-Ord-03	Approved	PUNITH K
BOOK-F-015	Real-time Order Tracking	OrderModule	TC-Ord-04	Approved	PUNITH K
BOOK-F-016	Customer Order History Log	OrderModule	TC-Ord-05	Approved	PUNITH K
BOOK-F-017	Catalog	AdminModule	TC-Adm-01	Approved	PURVITHA

Req ID	Requirement Summary	Design Module	Test Case Ref	Status	Owner / Assignee
	Curation & Approval				MACHIREDDY
BOOK-F-018	Inventory & Order Status Update	AdminModule	TC-Adm-02	Approved	PURVITHA MACHIREDDY
BOOK-F-019	Review Submission & Moderation	AdminModule	TC-Adm-03	Approved	PURVITHA MACHIREDDY
BOOK-F-020	Sales Dashboard Analytics	AdminModule	TC-Adm-04	Approved	PURVITHA MACHIREDDY
BOOK-NF-001	Response Latency (\le2s)	API/Database	TC-Perf-01	Approved	PURVITHA MACHIREDDY
BOOK-NF-002	Monthly Uptime (\ge99.5\%)	Infrastructure	TC-Rel-01	Approved	PURVITHA MACHIREDDY
BOOK-NF-003	Concurrency (500 Users)	Backend Server	TC-Perf-02	Approved	PURVITHA MACHIREDDY
BOOK-NF-004	Responsive Accessibility UI	React Frontend	TC-UX-01	Approved	PURVITHA MACHIREDDY
BOOK-NF-005	3-Year Order Data Retention	PostgreSQL DB	TC-Dat-01	Approved	PURVITHA MACHIREDDY
BOOK-NF-006	ACID Transaction Compliance	PostgreSQL DB	TC-Int-01	Approved	PURVITHA MACHIREDDY
BOOK-SR-001	Password Hashing (bcrypt)	AuthModule	TC-Sec-01	Approved	PRUTHVIRAJ H
BOOK-SR-002	TLS 1.2/1.3 Enforcement	Infrastructure	TC-Sec-02	Approved	PURVITHA MACHIREDDY
BOOK-SR-003	JWT Token & Revocation	AuthModule	TC-Sec-03	Approved	PRUTHVIRAJ H
BOOK-SR-004	Granular Server-Side RBAC	Admin/API	TC-Sec-04	Approved	PURVITHA MACHIREDDY
BOOK-SR-005	Payment Tokenization (PCI)	OrderModule	TC-Sec-05	Approved	PUNITH K

THANK YOU

-TEAM 4